

BOE 4.3 Version Upgrade

RDM DevOps is deploying BOE components to new BOE 4.3 Servers. URLs for respective environments are listed below for direct access to new BOE environments:

ENV	Launch Pad URL
DEV	https://boe.dev.int.ttc.ca/BOE/BI
POC	https://boe.poc.int.ttc.ca/BOE/BI
TST	https://boe.tst.int.ttc.ca/BOE/BI
UAT	https://boe.uat.int.ttc.ca/BOE/BI
PROD	https://boe.int.ttc.ca/BOE/BI

RDM DevOps is updated to deploy **only** to new BOE 4.3 Servers. Currently existing BOE components are preloaded to new servers to make sure you carry over existing permission to report folders and you can check and verify you are able to access folder. Folder content on new environment likely may not be the latest version and developer is responsible to deploy the latest version to new BOE servers once ready to upgrade.

Current BOE 4.2 servers and BOE API service are going to be active and running until all existing BOE components are migrated so your system is able to run without interruption until you upgrade to new BOE 4.3 environment.

Once you make change to your BOE components, you will have to update, and redeploy your BOE components to new BOE 4.3 servers. RDM DevOps is updated to deploy **only** to new BOE 4.3 Servers. All RDM DevOps Reports components BOE links, for all environments, open new launch pad URLs listed in table above. If you want to access 4.2 servers, you will have to type it in your address bar manually. For example, you need to type <http://leaskdale:8080/BOE/BI> because RDM DevOps links now open BOE 4.3 launch pad for dev environment, which is <https://boe.dev.int.ttc.ca/BOE/BI>.

HOW TO UPDATE BOE COMPONENTS

Step 1. Obtain required software on your laptop. Create Service Request to get it installed on your workstation:

- SAP Business Intelligence Client 4.3 (Note that existing 4.2 client needs to be uninstalled from your workstation first, and workstation needs to reboot before installing new version)
- SAP Crystal Reports 2020 version
- (optional) if using Oracle database ODBC, you need to get Oracle 64-bit client installed on your workstation

Step 2. Create 64-bit ODBC connection to work with new BOE 4.3 environment and Crystal Reports 2020 as all ODBC connections needs to be set as 64-bit, for both, SQL Server or Oracle database connection types.

Step 3. Contact your **BOE Administrator** to see if you need to manually reload your Business View or Universe component out of your component **export.lcmbar** file, which is checked in your component source control folder. Most of business views are preloaded to BOE dev environment as the component is created manually in dev environment. However, it may not be up to date with latest changes, so it is the developer's responsibility to make sure it is up to date.

Step 4. Crystal Reports files may need to be updated to latest version, if very old version file is saved in source control, or if report has embedded printer inside report properties. 'No printer' option should be selected for each report file.

Step 5. Create Change Order to deploy all of your BOE components and dependent Business Views to make sure that your version is up to date on BOE 4.3 servers.

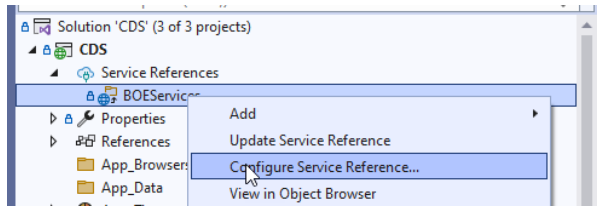
HOW TO UPDATE WEB SERVICE REFERENCE

If your application is referencing web service to communicate with BOE servers from your application code, you will need to update your web application and deploy along with all of your BOE components once you make your next change to BOE components. **Please contact BOE Administrator to update BOE password** for your application and update your configuration file for each environment. Make sure you encrypt the changed value, if you are using encryption. Default password (no encryption in this example) is set as "Default!@USERid12" or if your code is using encrypted value then "zHLg3GZE2uCITfeiB9JGX2uFuLoyP6+RmITFuXiTbLk=". Other login parameters and BOE folder names are unchanged.

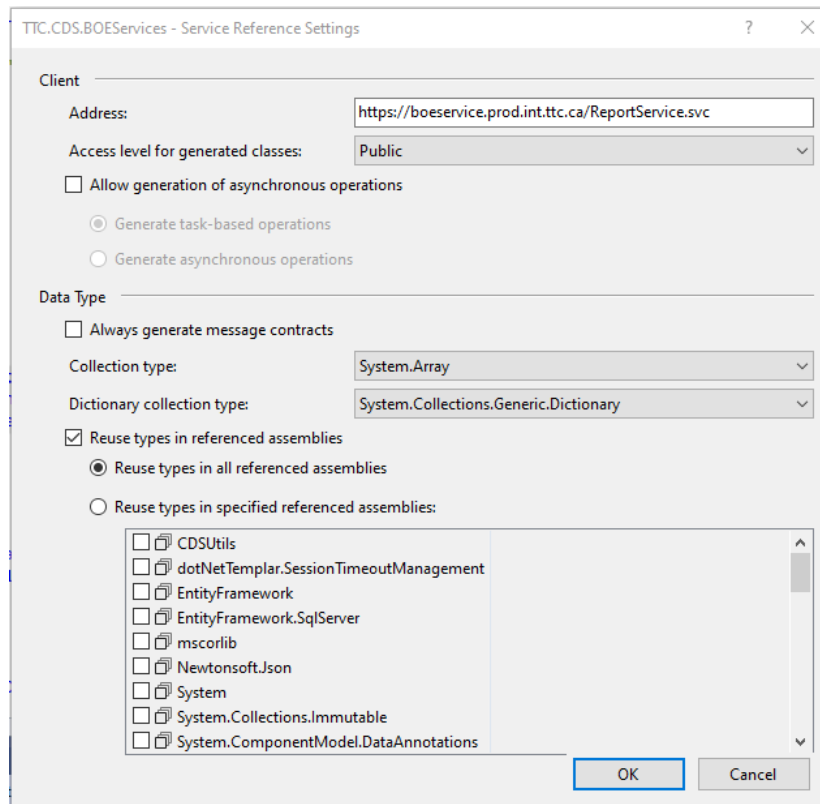
BOE 4.3 environment requires 64-bit ODBC connections when editing and changing report so if you are using Oracle database connection, you will need to request Oracle 19 64-bit client, SAP Business Intelligence Client 4.3 version and Crystal Reports 2020 version. You will need to submit service request to EUD to update your workstation with above listed installations.

Steps to update .net 4.8 Framework application web service reference to work with new BOE 4.3 environments

Step 1. Right click on Service References and click Configure Service Reference...



Update your service reference to: <https://boeservice.prod.int.ttc.ca/ReportService.svc>

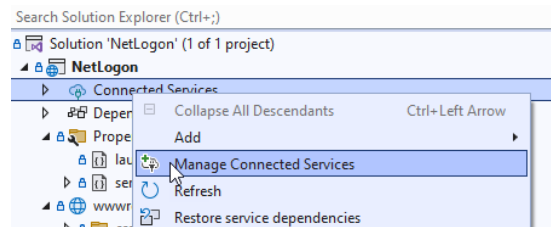


Step 2. Update your web.config files for all environments to reflect this change as your local web config updated when updated service reference. For example:

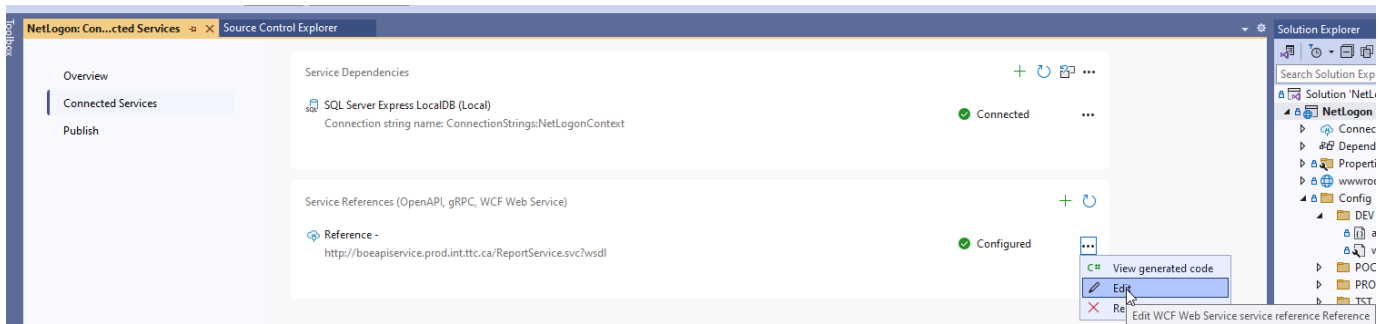
```
<system.serviceModel>
  <bindings>
    <basicHttpBinding>
      <binding name="BasicHttpBinding_IReportService">
        <security mode="Transport" />
      </binding>
    </basicHttpBinding>
  </bindings>
  <client>
    <endpoint address="https://boeservice.prod.int.ttc.ca/ReportService.svc/soap"
      binding="basicHttpBinding" bindingConfiguration="BasicHttpBinding_IReportService"
      contract="BOEServiceProxy.IReportService" name="BasicHttpBinding_IReportService" />
  </client>
</system.serviceModel>
```

Steps to add/update .NET Core Framework application web service reference to work with new BOE 4.3 environments

Step 1. Right click on Service References and click Manage Connected Services

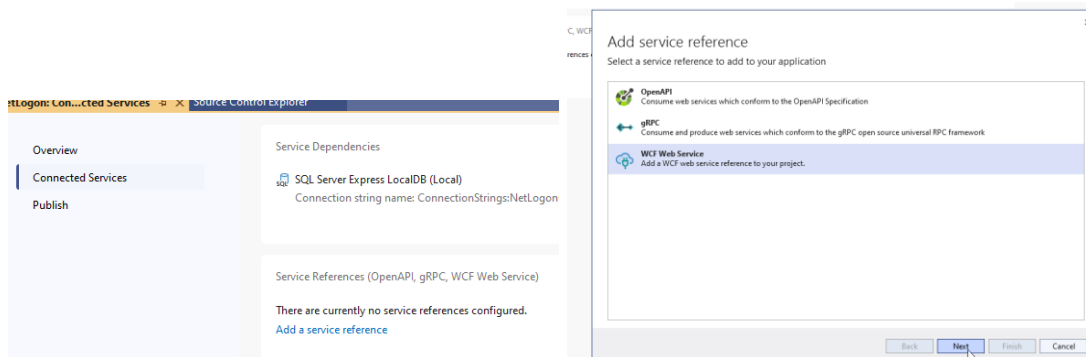
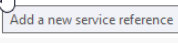


To Edit existing web service reference, click Edit as on the screenshot. If Edit values are disabled, you can remove existing reference and follow 'Add' steps.

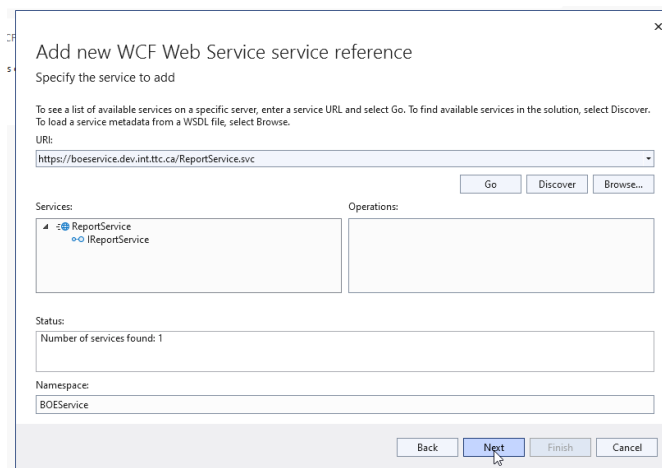


Step 2. Click on Add a service reference if you are adding it

Add a service reference



Step 3. Insert or update your service reference to: <https://boeservice.prod.int.ttc.ca/ReportService.svc>
Make sure you reference production web service link above (not dev or another test domain).



Click Next.

Step 4. Check the settings as on the screenshot below and click Next.

The screenshot shows the 'Add new WCF Web Service service reference' dialog box with the 'Specify the data type options' tab selected. The 'Always generate message contracts' checkbox is unchecked. The 'Collection type' is set to 'System.Array' and the 'Dictionary collection type' is set to 'System.Collections.Generic.Dictionary'. The 'Reuse types in referenced assemblies' checkbox is checked, with the radio button for 'Reuse types in all referenced assemblies' selected. A list of assemblies is shown below, including 'dotnet-aspnet-codegenerator-design', 'Microsoft.AspNetCore.Html.Abstractions', 'Microsoft.AspNetCore.Razor', 'Microsoft.AspNetCore.Razor.Language', 'Microsoft.AspNetCore.Razor.Runtime', 'Microsoft.Bcl.AsyncInterfaces', 'Microsoft.CodeAnalysis', 'Microsoft.CodeAnalysis.CSharp', and 'Microsoft.CodeAnalysis.CSharp.Workspaces'. The 'Next' button is highlighted with a mouse cursor.

Step 5. Check Generate Synchronous Operations and click Finish.

The screenshot shows the 'Add new WCF Web Service service reference' dialog box with the 'Specify the client options' tab selected. The 'Access level for generated classes' is set to 'Public'. The 'Generate Synchronous Operations' checkbox is checked, and the 'Enable Data Binding' checkbox is unchecked. The 'Finish' button is highlighted with a mouse cursor.

Step 6. Verify ConnectedService.json contains reference to the web service <https://boeservice.prod.int.ttc.ca/ReportService.svc> and Reference.cs contains corresponding service model

The screenshot shows the Visual Studio editor with the 'ConnectedService.json' file open. The file contains a JSON object with the following properties: 'ExtendedData' (an object with 'inputs' and 'collectionTypes' arrays), 'namespaceMappings' (an array with a single entry '*, BOEService'), 'sync' (true), 'targetFramework' ('net8.0'), and 'typeReuseMode' ('All'). The 'inputs' array contains the URL 'https://boeservice.prod.int.ttc.ca/ReportService.svc?wsdl'. The 'collectionTypes' array contains 'System.Array' and 'System.Collections.Generic.Dictionary`2'. The 'namespaceMappings' array contains a single entry '*, BOEService'. The 'sync' property is true, 'targetFramework' is 'net8.0', and 'typeReuseMode' is 'All'.